

Bauchi State University, Gadau

Faculty of Science – Department of Physics

Course Code: PHY1201 – Mechanics

Academic Session: 2020/2021 and 2023/2024 – First Semester Examination

Time Allowed: 45 Minutes

Instruction: ANSWER ALL QUESTIONS

1. If three forces acting at a point are in equilibrium, each force is proportional to the sine of the angle between the other two. This is called:

- (a) Parallelogram Law
- (b) Triangle of force
- (c) Pythagoras theorem
- (d) Lami's theorem ☒

2. Convert the system of units to another:

- (a) Check the correctness of the physical equations
- (b) Check the dimension of the quantities ☒
- (c) Derive the important relations
- (d) None of the above

3. The branch of mechanics that deals with forces acting on systems not experiencing acceleration is:

- (a) Dynamics
- (b) Kinematics
- (c) Statics ☒
- (d) Mechanics

4. An ice skater propels herself with a posterior force of 30N and lateral force of 40N. What is the resultant force?

- (a) 64.3 N ☒
- (b) 40.63 N
- (c) 34.66 N
- (d) 50 N

5. Length, mass, and time are base quantities because:

- (a) They cannot be derived from other quantities ☒
- (b) They can be derived
- (c) They depend on fundamental units
- (d) They partially depend

6. Dimensions of strain are:

- (a) [L]
- (b) $[ML^{-1}T^{-2}]$
- (c) $[L^{-1}]$
- (d) Dimensionless quantity ☒

7. The branch of mechanics dealing with forces on systems not accelerating is:

- (a) Dynamics
- (b) Kinematics
- (c) Statics ☒
- (d) Mechanics

8. A number used to describe a physical phenomenon quantitatively is called:

- (a) Quantity
- (b) Reference standard
- (c) Unit
- (d) Physical quantity ☒

9. According to vector algebra:

- (a) $m(A + B) = mA + mB$ ☒
- (b) Commutative law for addition
- (c) Associate law
- (d) Commutative law for multiplication

10. All are characteristics of standard units except:

- (a) Well defined (e.g., cubit) ☒
- (b) Convenient size
- (c) Easily produced
- (d) Reproducible

11. Centripetal acceleration is expressed as:

- (a) $a = v^2/r$ ☒
- (b) $a = r^2/v$
- (c) $a = v/r^2$
- (d) $a = r/v^2$

12. Angular speed (ω) is:

- (a) $\omega = L/I$
- (b) $\omega = kx$
- (c) $\omega = r\theta$
- (d) $\omega = \theta/t$ ☒

13. Relationship between linear velocity and angular speed:

- (a) $v = \omega r$ ☒
- (b) $v = \theta t$
- (c) $v = \omega/t$
- (d) $v = \omega\theta$

14. Nine complete revolutions in radians:

- (a) 14π
- (b) 18π ☒

- (c) 9π
- (d) 20π

15. A vehicle with wheel radius 0.6 m moving at 20 m/s. Wheel rotation rate?

- (a) 20 r/s
- (b) 40 r/s
- (c) 10 r/s ☒
- (d) 30 r/s

16. When all particles of a body are concentrated, it refers to:

- (a) Momentum
- (b) Moment of inertia
- (c) Mass ☒
- (d) Force

17. Moment of inertia for a rod of length l and mass m about one end:

- (a) $I = ml^2/3$
- (b) $I = 2mr^2$
- (c) $I = mr^2 \sin\theta$
- (d) $I = ml^2/12$ ☒

18. Angular displacement (θ) is defined as:

- (a) $s = rP\sin\theta$
- (b) $s = sr$
- (c) $\theta = s/r$ ☒
- (d) $v = s/t$

19. Change in angular velocity with time is:

- (a) Circular displacement
- (b) Angular acceleration ☒
- (c) Angular speed
- (d) Force

20. Kepler's third law states:

- (a) $r^2 \propto r^3$
- (b) $T^2 \propto r^3$ ☒
- (c) $T \geq r$
- (d) $T = r$

21. Equation $(p + a/v^2)(v - b) = \text{constant}$. Unit of a ?

- (a) Dyne $\times \text{cm}^2$
- (b) Dyne $\times \text{cm}^4$ ☒
- (c) Dyne $\times \text{cm}^3$
- (d) Dyne $\times \text{cm}^{-1}$

22. Force $F = at + bt^2$. Dimensions of a and b?

- (a) MLT^{-3} and MLT^{-2} ✓
- (b) MLT^{-1} and MLT^{-2}
- (c) MLT^{-1} and MLT^0
- (d) MLT^{-2} and MLT^{-3}

23. Example of scalar quantity:

- (a) Velocity
- (b) Force
- (c) Volume ✓
- (d) Acceleration

24. Car starts from rest, accelerates over 5s for 110m. Acceleration?

- (a) 8.10 m/s^2 ✓
- (b) 8.5 m/s^2
- (c) 8.0 m/s^2
- (d) 8.8 m/s^2

25. Displacement of a team running 20km and returning to stadium?

- (a) Stadium
- (b) Zero ✓
- (c) 20km
- (d) None

26. Work done = 3 KJ, distance = 5 Km, time = 30s. Velocity?

- (a) 16.66 Km/s
- (b) 166.7 m/s
- (c) 15 hrs
- (d) 6 m/s ✓

27. Mass of man if velocity = 10 m/s?

- (a) 2.159 Kg ✓
- (b) 2.159 g
- (c) 0.2159 Kg
- (d) 10 Kg

28. Which does NOT represent Newton's 2nd law?

- (a) $F = ma$
- (b) $Ft = mv - mu$
- (c) Impulse = momentum change
- (d) $v^2 = u^2 + 2as$ ✓

29. Body projected upward at 20 m/s at 30° . Time to reach ground?

- (a) 20 sec ✓
- (b) 10 sec

- (c) 30 mins
- (d) 30 sec

30. Inverse square law:

- (a) $F = ma$
- (b) $F = GMm/r^2$ ✓
- (c) $V = \sqrt{3Gr}$
- (d) None

31. 1.0 kg block on cord around wheel ($r = 0.3\text{m}$), covers 2.0m in 5s. Moment of inertia?

- (a) 21.5 m/s^2
- (b) 2.5 m/s^2
- (c) $22.4 \text{ kg}\cdot\text{m}^2$ ✓
- (d) 25.5 m/s^2

32. Earth-moon distance = $3.8 \times 10^8\text{m}$. $M_e = 6 \times 10^{24} \text{ Kg}$, $M_m = 7.4 \times 10^{22} \text{ Kg}$. Center of mass?

- (a) $8.6 \times 10^6 \text{ m}$
- (b) $1.6 \times 10^6 \text{ m}$ ✓
- (c) $1 \times 10^6 \text{ m}$
- (d) $6.0 \times 10^6 \text{ m}$

33. Three equal masses form equilateral triangle of 1.0m sides. Center of mass?

- (a) 0.91
- (b) 0.11, 5.1L
- (c) 0.1L
- (d) 0.5L, 0.29L ✓

34. Velocity in SHM:

- (a) $V = \omega\sqrt{r^2/y^2}$
- (b) $V = \omega\sqrt{r^2 + y^2}$
- (c) $V = \sqrt{r^2 - y^2}$
- (d) $V = \omega\sqrt{r^2 - y^2}$ ✓

35. Simple pendulum length = 0.36m, released at 10° . Speed of bob?

- (a) 6.2 m/s
- (b) 0.2 m/s
- (c) 4.2 m/s
- (d) 0.3 m/s ✓

Here's the continuation with Questions 36–60 from the PHY1201 Mechanics test at Bauchi State University, Gadau — written clearly and with correct answers ticked ✓.

36. A bullet is fired horizontally with a velocity of 60 m/s from the top of a building 70 m high. How far from the foot of the building will the bullet land?

- (a) 227 cm
- (b) 227 m ☒
- (c) 0.62 m
- (d) 0.62 cm

37. Earth exerts a force of about 800 N on each person standing on it. What is the approximate mass of the person?

- (a) 800 kg
- (b) 800 kgm
- (c) 80 kg ☒
- (d) 80 kgm

38. In an elastic collision, all energy remains as kinetic energy. No energy is lost to other forms. This is called:

- (a) Elastic collision
- (b) Inelastic collision
- (c) Perfectly inelastic collision
- (d) Perfectly elastic collision ☒

39. If $\theta = 90^\circ$, the work done will be:

- (a) Positive work
- (b) Zero work ☒
- (c) Simple work
- (d) Negative work

40. Two people of equal mass (100 kg) stand 20 m apart. What is the gravitational force between them?

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

- (a) $1.85 \times 10^{-6} \text{ N}$
- (b) $1.85 \times 10^{-8} \text{ N}$ ☒
- (c) $1.85 \times 10^{-10} \text{ N}$
- (d) $1.85 \times 10^{-4} \text{ N}$

41. A uniform meter rule is balanced at the 20 cm mark when a 20 N weight hangs at that point. What is the weight of the meter rule?

- (a) 10 N
- (b) 12 N
- (c) 2 N ☒
- (d) 1 N

42. Kinematics deals with describing motion without reference to:

- (a) Force ☒
- (b) Velocity
- (c) Acceleration
- (d) Displacement

43. Physical quantities can be classified based on their properties or laws as:

- (a) Fundamental quantities
- (b) Derived quantities
- (c) Scalar and vector quantities ☒
- (d) None of the above

44. The value of gravitational constant G in MKS system is:

- (a) $6.67 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$ ☒
- (b) $6.67 \times 10^{-11} \text{ m.k.s units}$
- (c) $6.67 \times 10^{-11} \text{ m.k.s unit}$
- (d) $6.67 \times 10^{-11} \text{ cm}^3\text{kg}^{-1}\text{s}^{-2}$

45. Dimensional formula of G is:

- (a) $[M^{-1}L^3T^{-2}]$ ☒
- (b) $[ML^{-3}T^2]$
- (c) $[MLT^{-2}]$
- (d) $[M^{-1}L^3T^{-2}]$

46. Which of the following is NOT valid for the position of a car at time t starting from rest?

- (a) $x = ut + \frac{1}{2}at^2$
- (b) $x = vt + \frac{1}{2}at^2$ ☒
- (c) $x = \frac{1}{2}at^2$
- (d) $x = ut$

47. Dimensional formula of energy is:

- (a) $[ML^2T^{-2}]$ ☒
- (b) $[MLT^{-2}]$
- (c) $[ML^2T^2]$
- (d) $[MLT]$

48. Power = 96 watts, time = 24 s, force = 4 N. Find power.

- (a) 3 watts ☒
- (b) 2 watts
- (c) 1 watt
- (d) 96 watts

49. Unit vector in direction of $A = 3i + 4j$ is:

- (a) $(3i + 4j)/5$ ☒
- (b) $(3i + 4j)/12$
- (c) $(3i + 4j)/13$
- (d) $(3i + 4j)/25$

50. Relationship between linear and angular velocity:

- (a) $v = r\omega$, measured in m/s ☒
- (b) $v = r$, measured in m/s

- (c) $v = r\omega$, measured in m
- (d) $v = r\omega$, measured in r

51. A wheel turns at 2 m/s and coasts to rest in 30 s. How far does it turn?

- (a) 60 m
- (b) 30 m
- (c) 180 m ☒
- (d) 60π rad

52. Horizontal component of radial acceleration:

- (a) $a = v^2/r$ ☒

53. Moment of inertia of a uniform meter rule rotating about its center (mass = 0.5 kg, length = 1 m):

- (a) $0.1 \text{ kg}\cdot\text{m}^2$
- (b) 0.25 kg
- (c) 0.025 m
- (d) $0.025 \text{ kg}\cdot\text{m}^2$ ☒

54. Moment of inertia of a sphere:

- (a) $I = ma^2$
- (b) $I = \frac{1}{2}ma^2$
- (c) $I = \frac{2}{3}mr^2$ ☒
- (d) $I = \frac{1}{2}mr^2$

55. Moment of inertia of a spherical ball bearing (mass = 0.5 kg, diameter = 0.2 cm):

- (a) $0.02 \text{ kg}\cdot\text{cm}^2$
- (b) $0.02 \text{ kg}\cdot\text{cm}^2$
- (c) $0.002 \text{ kg}\cdot\text{cm}^2$ ☒
- (d) 0.05 kg^2

56. Which of the following is a scalar quantity?

- (a) Displacement
- (b) Velocity
- (c) Speed ☒
- (d) Force

57. Which of the following is a vector quantity?

- (a) Mass
- (b) Temperature
- (c) Acceleration ☒

(d) Energy

58. Which of the following is a derived quantity?

- (a) Length

- (b) Mass
- (c) Time
- (d) Force ☒

59. Which of the following is a fundamental quantity?

- (a) Force
- (b) Pressure
- (c) Time ☒
- (d) Velocity

60. Which of the following is NOT a base quantity in mechanics?

- (a) Length
- (b) Mass
- (c) Time
- (d) Temperature ☒